

1.2 – Purpose of the Project

The Visualization Tool for Electric Vehicle (EV) Charge and Range Analysis, built with Tableau helps people understand and work with EV charging and range data in a way.

This tool shows how things like battery size charging time energy use and driving conditions affect how well an EV works.

- The tool turns raw EV data into charts and dashboards.
- This lets users keep an eye on their battery health see how well their EV charges compare their vehicles performance to others and spot trends in energy use.
- It is useful for EV owners, fleet managers, researchers and organizations that want to make their vehicle operations better and more sustainable.

The main goals of the tool are:

- To track how people, charge their EVs and use their batteries.
- To study how far EVs can go on a charge and how efficiently they use energy.
- To compare the performance of EV models.
- To find out what affects battery performance and range.
- To help manage charging in a way and save energy.
- To provide insights through interactive Tableau visualizations.
- Overall the project helps people understand EV performance better.
- It promotes using data to make EVs more efficient, reliable and environmentally friendly.
- The Visualization Tool for Electric Vehicle (EV) Charge and Range Analysis supports transportation.
- It helps EV owners and organizations make decisions, about their EV operations.